

EISENSTEIN PRIMES FOR $\Gamma_0(3)$: AN EXPLICIT CONSTANT-TERM COMPUTATION

ABSTRACT. For even weight $k \geq 4$ and primes $q \neq 2, 3$, we show that q is an Eisenstein prime of $S_k(\Gamma_0(3))$ if and only if q divides the numerator of $\frac{(3^k-1)B_k}{2k}$. The argument is an instance of the Eisenstein-ideal theory of Mazur and Ribet: its only nonformal input is the explicit constant term, at the width-three cusp, of the eigenform $\Phi_k = G_k - G_k|V_3$, where G_k is the normalized level-one Eisenstein series. We give that computation in full, note that it accounts simultaneously for the level-three factor $3^k - 1$ and the Bernoulli factor B_k , and we record the precise limits of the statement: multiplicities and the primes 2, 3 are not treated. The prime-set statement was found empirically and verified by an independent Hecke-determinant computation; this note supplies the proof of that empirical observation.

1. INTRODUCTION

Eisenstein congruences—congruences between cusp forms and Eisenstein series—are governed, at prime level, by constant terms of Eisenstein series, which are essentially (generalized) Bernoulli numbers. This is the Eisenstein-ideal theory of Mazur [3] and Ribet [4]. The present note treats one explicit case, the group $\Gamma_0(3)$, for which a precise characterization of the relevant primes was first observed computationally. We supply the one analytic step that pins the characterization down—the constant term of the relevant eigenform at the cusp of width three—and thereby promote the empirical observation to a theorem. We claim no new method; the interest is in the explicit level-three formula and in the fact that a single constant term carries *both* the level factor $3^k - 1$ and the level-one Bernoulli factor.

2. NOTATION

Let $k \geq 4$ be even and let

$$G_k(\tau) = -\frac{B_k}{2k} + \sum_{n \geq 1} \sigma_{k-1}(n)q^n, \quad q = e^{2\pi i\tau},$$

be the weight- k Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$ normalized so that $a_1(G_k) = 1$; thus $a_0(G_k) = -B_k/2k$ and $G_k(-1/\tau) = \tau^k G_k(\tau)$. Write V_3 for the operator $(V_3 g)(\tau) = g(3\tau)$ and set

$$\Phi_k(\tau) := G_k(\tau) - G_k(3\tau) \in M_k(\Gamma_0(3)).$$

The two cusps of $\Gamma_0(3)$ are ∞ (width 1) and 0 (width 3); we write c_∞ and c_0 for the corresponding constant terms, the latter computed after slashing by a scaling matrix for the cusp 0.

By an *Eisenstein prime* of $S_k(\Gamma_0(3))$ we mean a prime q for which there is a cuspidal Hecke eigenform $f \in S_k(\Gamma_0(3))$ with

$$a_f(p) \equiv 1 + p^{k-1} \pmod{q} \quad \text{for all primes } p \neq 3.$$

3. MAIN THEOREM

Theorem 1. *Let $k \geq 4$ be even and let q be a prime with $q \neq 2, 3$. Then q is an Eisenstein prime of $S_k(\Gamma_0(3))$ if and only if q divides the numerator of $\frac{(3^k-1)B_k}{2k}$.*

4. PROOF

Step 1: the eigenform and its eigenvalues. From the divisor-sum identity

$$(1) \quad \sigma_{k-1}(3n) = (1 + 3^{k-1})\sigma_{k-1}(n) - 3^{k-1}\sigma_{k-1}(n/3)$$

(where $\sigma_{k-1}(n/3) = 0$ if $3 \nmid n$) one checks directly that the integral q -expansion $a_{\Phi_k}(n) = \sigma_{k-1}(n) - \sigma_{k-1}(n/3)$, with $a_{\Phi_k}(0) = 0$ and $a_{\Phi_k}(1) = 1$, satisfies

$$T_p \Phi_k = (1 + p^{k-1})\Phi_k \quad (p \neq 3), \quad U_3 \Phi_k = 3^{k-1}\Phi_k.$$

Indeed, for $p \neq 3$ the Hecke relation $a(pn) + p^{k-1}a(n/p) = (1 + p^{k-1})a(n)$ for $a = a_{\Phi_k}$ follows from (1) applied to each summand, and $a_{\Phi_k}(3n) = 3^{k-1}a_{\Phi_k}(n)$ follows from (1) directly. Thus Φ_k is an Eisenstein eigenform carrying the eigenvalue system $1 + p^{k-1}$ ($p \neq 3$).

The Eisenstein subspace of $M_k(\Gamma_0(3))$ is two-dimensional, spanned by $G_k(\tau)$ and $G_k(3\tau)$; both are fixed by every T_p ($p \neq 3$) with eigenvalue $1 + p^{k-1}$, so the prime-to-3 Hecke eigenvalue system $1 + p^{k-1}$ does not distinguish them. The operator U_3 does: in the basis $(G_k(\tau), G_k(3\tau))$ one computes

$$U_3 = \begin{pmatrix} 1 + 3^{k-1} & 1 \\ -3^{k-1} & 0 \end{pmatrix},$$

with eigenvalues 1 and 3^{k-1} . The eigenvalue 3^{k-1} belongs to $\Phi_k = G_k(\tau) - G_k(3\tau)$, and the eigenvalue 1 belongs to $\Psi_k := G_k(\tau) - 3^{k-1}G_k(3\tau)$.

Step 2: the constant terms of Φ_k . At ∞ ,

$$c_\infty(\Phi_k) = a_0(G_k) - a_0(G_k) = -\frac{B_k}{2k} + \frac{B_k}{2k} = 0.$$

At the cusp 0 we slash by $\sigma_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, so $(g|_k \sigma_0)(\tau) = \tau^{-k}g(-1/\tau)$. Since G_k is $\mathrm{SL}_2(\mathbb{Z})$ -invariant, $G_k|_k \sigma_0 = G_k$ and hence $c_0(G_k) = -B_k/2k$. For the translate, writing $w = \tau/3$,

$$G_k(3 \cdot (-1/\tau)) = G_k(-1/w) = w^k G_k(w) = (\tau/3)^k G_k(\tau/3),$$

so $(V_3 G_k)|_k \sigma_0(\tau) = \tau^{-k}(\tau/3)^k G_k(\tau/3) = 3^{-k}G_k(\tau/3)$, whose constant term is $3^{-k}(-B_k/2k)$. Therefore

$$(2) \quad c_0(\Phi_k) = -\frac{B_k}{2k} - 3^{-k} \left(-\frac{B_k}{2k} \right) = -\frac{B_k}{2k} \cdot \frac{3^k - 1}{3^k}.$$

For a prime $q \neq 3$ the factor 3^{-k} is a unit, so the q -adic valuation of (2) equals that of $\frac{(3^k-1)B_k}{2k}$; in particular q divides the numerator of $c_0(\Phi_k)$ if and only if q divides the numerator of $\frac{(3^k-1)B_k}{2k}$.

Step 3: the congruence. ($\Leftarrow$) Suppose $q \mid \mathrm{num}\left(\frac{(3^k-1)B_k}{2k}\right)$, $q \neq 2, 3$. The form Φ_k has q -integral Fourier coefficients (they are integers), and by Step 2 both of its constant terms vanish modulo q : the one at ∞ is 0, and the one at 0 has numerator divisible by q . By the q -expansion principle [2, 1], a q -integral form on $\Gamma_0(3)$ whose constant terms vanish modulo q at every cusp reduces modulo q to a cusp form. Hence $\Phi_k \bmod q$ lies in the space of mod- q cusp forms and carries the Hecke eigenvalues $1 + p^{k-1}$ ($p \neq 3$); by the Deligne–Serre lifting lemma there is a cuspidal eigenform $f \in S_k(\Gamma_0(3))$ with $a_f(p) \equiv 1 + p^{k-1} \pmod{q}$ for all $p \neq 3$. Thus q is an Eisenstein prime.

($\Rightarrow$) Suppose $q \neq 2, 3$ is an Eisenstein prime, witnessed by a cuspidal eigenform f with $a_f(p) \equiv 1 + p^{k-1} \pmod{q}$ for all $p \neq 3$. Reducing the boundary (constant-term) exact sequence modulo q , such a congruence between f and the Eisenstein subspace is realized by an Eisenstein eigenform sharing the eigenvalues of f at all Hecke operators, including U_3 . The two Eisenstein eigenforms are Φ_k ($U_3 = 3^{k-1}$) and Ψ_k ($U_3 = 1$) from Step 1, so f is congruent modulo q to one of them, and matching the (vanishing) constant terms of f forces q to divide the corresponding numerator. For Φ_k this numerator is that of $\frac{(3^k-1)B_k}{2k}$ by (2). For Ψ_k a parallel computation gives

$$c_\infty(\Psi_k) = -\frac{B_k}{2k}(1 - 3^{k-1}), \quad c_0(\Psi_k) = -\frac{B_k}{2k}(1 - 3^{-1}) = -\frac{B_k}{3k},$$

so the congruence via Ψ_k forces $q \mid \text{num}(B_k/2k)$. Since $\frac{(3^k-1)B_k}{2k} = (3^k-1) \cdot \frac{B_k}{2k}$ with $3^k - 1 \in \mathbb{Z}$, every prime $q \neq 3$ dividing $\text{num}(B_k/2k)$ also divides $\text{num}(\frac{(3^k-1)B_k}{2k})$. In either case $q \mid \text{num}(\frac{(3^k-1)B_k}{2k})$, as required. $\square$

5. COMPUTATIONAL VERIFICATION

The following were checked by exact `Python/SymPy` computation.

[leftmargin=2em]The eigenform identities $T_p\Phi_k = (1 + p^{k-1})\Phi_k$ ($p \in \{2, 5, 7, 11, 13\}$) and $U_3\Phi_k = 3^{k-1}\Phi_k$ were verified on Fourier coefficients a_n , $n < 40$, for all even k with $6 \leq k \leq 60$. The identity of numerators in (2), namely $\text{num}(c_0(\Phi_k)) = \text{num}(\frac{(3^k-1)B_k}{2k})$ up to powers of 3, was verified for all even k with $4 \leq k \leq 60$. For $k = 6, 12, 18, 24, 30, 36$ the large prime factors (> 11) of $\text{num}(\frac{(3^k-1)B_k}{2k})$ were compared with the primes dividing the Hecke determinant $\gcd_p \det(T_p - (1 + p^{k-1})I)$ on $S_k(\Gamma_0(3))$ (with $p \in \{2, 5, 7, 11, 13\}$); the two sets coincide in every case.

We note one cautionary data point consistent with the scope of Theorem 1: at $k = 54$ the small prime 23 appeared in the five-prime determinant gcd but is *not* an Eisenstein prime—enlarging the Hecke set removes it, and the direct mod-23 test shows the matrices $T_p - (1 + p^{53})I$ have trivial common kernel. This is an artifact of the finite-Hecke-set gcd, not a counterexample; it illustrates why the gcd diagnoses the prime set only after enough Hecke operators are used.

6. SCOPE AND LIMITATIONS

Theorem 1 determines the *set* of Eisenstein primes for $q \neq 2, 3$. It does not determine *multiplicities*: when $\det(T_p - (1 + p^{k-1})I)$ is divisible by a higher power of q , controlling that power requires a finer analysis (deformation rings, or the ordinary filtration) not carried out here. The primes $q = 2$ and $q = 3$ are excluded: $q = 3$ is the level, and $q = 2$ requires separate treatment of the relevant denominators. Finally, the empirical multiplicity-2 behavior of the largest prime observed in several weights is *not* explained by the present argument and is recorded only as an open question.

Remark 2. The level $\Gamma_0(3)$ is attached to the elliptic curve $y^2 = x^3 + 1$, which has complex multiplication by the Eisenstein integers; its supersingular primes are exactly those $\equiv 2 \pmod{3}$, and by Ogg's observation the primes p for which all supersingular j -invariants are rational over $\mathbb{F}_p$ are the prime divisors of the order of the Monster. The Bernoulli arithmetic appearing here is of a different nature (it governs Eisenstein congruences, not supersingular reduction); we mention the coincidence of level only, and claim no structural link.

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